

Shushmita Tiwari | Bangalore, India | [Portfolio](#) | [Github](#) | [Linkedin](#)

EDUCATION

Cambridge Institute of Technology, Bangalore (2023-2027)

Bachelor's Degree in Engineering (Artificial Intelligence & Machine Learning) | **CGPA : 9.00**

EXPERIENCE

Powerup Money | AI Engineer Intern | June 2026 – September 2026

- Architected a production LLM agent (LangGraph, FastAPI, Gemini) with dynamic context engineering, lazy retrieval, real-time SSE token streaming, and a 4-tier model fallback chain to minimize latency and maintain uptime.
- Independently designed and deployed a new specialized AI agent into a live production system defining its decision logic, prompt architecture, and tool integrations from the ground up.

ISRO | Machine Learning Intern | Nov 2025 – Feb 2026

- Developed and deployed production-ready ML pipelines for flood-prone area detection handling large-scale geospatial data ingestion, feature engineering, and model refinement for real-world application deployment and production support.
- Collaborated with scientists to translate complex problems into scalable AI/ML solutions, demonstrated ownership, analytical thinking, and ability to work across cross-functional teams with extended timelines and shifting priorities.

PROJECTS

SaffronAI – Voice-Based AI Restaurant Reservation Agent

Tech: Pipecat, Deepgram, Cartesia, Groq, Python, MongoDB, WebSocket

- Built a live, voice-based conversational AI agent (Pipecat, Deepgram, Cartesia, Groq) with sub-800ms round-trip latency, orchestrating real-time STT, LLM reasoning, and TTS into a structured multi-turn dialogue system architecturally mirroring production voice-interview agent design.
- Engineered custom conversational guardrails a frame-level processor enforcing single-question turn-taking and structured slot-filling before tool execution plus race-condition-safe tool calling, ensuring the agent never acts on unconfirmed or incomplete information.

SusJobs – ML-Based Fake Job Detection System

Tech: Python, Flask, Scikit-learn, Gemini API, TF-IDF, REST APIs

- Built and deployed a scalable ML pipeline with Flask REST API across 17K+ records; achieved 25% F1-score improvement through feature engineering demonstrating end-to-end application development and deployment ownership.
- Integrated Gemini LLM as a reasoning layer for interpretable predictions; managed application performance, latency, and reliability concerns in a production-style environment.

TECHNICAL SKILLS

AI/LLM: LangGraph, Gemini API, OpenAI API, Groq, Multi-Agent Systems, RAG Pipelines, Prompt Engineering, Pipecat, Deepgram, Cartesia (Voice AI)

Backend: Python, Java, FastAPI, Flask, Node.js/Express.js, REST APIs, JWT, OAuth 2.0, WebSocket

Database & Tools: MongoDB, Git/GitHub, React.js